Project Title: Musical Instrument Recognition Using DSP and ML

GROUP DETAILS :

Group Number – 4 (SECTION – B1)

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Synopsis

Problem Statement :

To develop a DSP and ML-based system for musical instrument recognition, covering both predominant-instrument classification and multilabel recognition of multiple instruments in musical audio recordings.

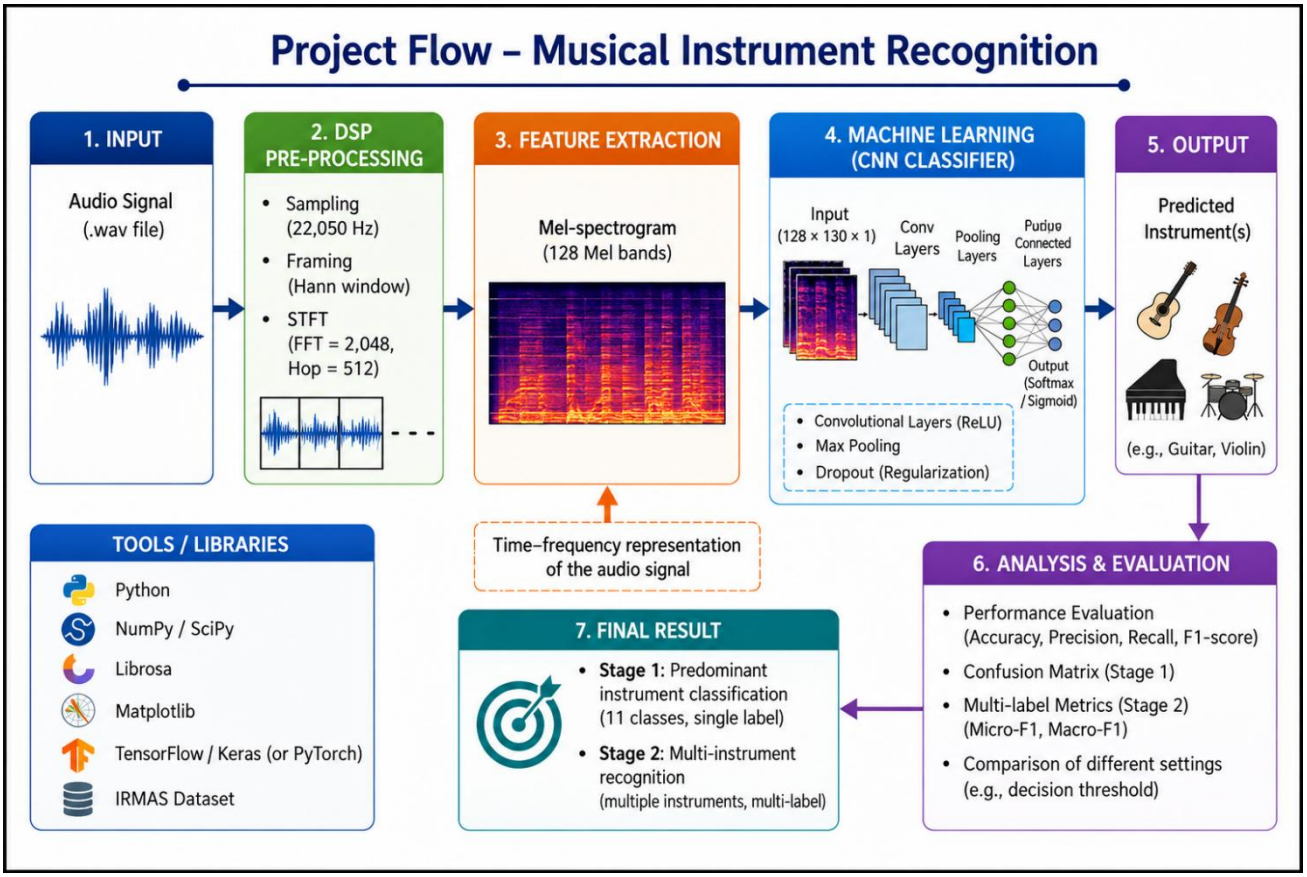
Introduction :

Musical instrument recognition is an important application of Digital Signal Processing (DSP) and Machine Learning (ML). Automatic identification is challenging because musical signals contain complex frequency patterns and harmonics, while recordings may include several instruments playing simultaneously.

This project will develop a system that processes musical audio and first predicts its predominant instrument, then extends recognition to multiple instruments. DSP techniques will generate informative audio features, which will be supplied to convolutional neural networks (CNNs) for classification.

Research Papers Referred :

1. Padmesh Sivalingam, Aamith Kishore T. J., Sri Krishna P., Yaswanth Reddy B., Ragav S., and Lekshmi C. R., “Robust CNN-based Musical Instrument Recognition with Enhanced Feature Learning,” *Proceedings of the International Conference on Inventive Computation Technologies (ICICT)*, pp. 851–857, **2025**. DOI: 10.1109/ICICT64420.2025.11005342.
2. Jiaxiang Zheng, Moxi Cao, and Chongbin Zhang, “ICKAN: A Deep Musical Instrument Classification Model Incorporating Kolmogorov-Arnold Network,” *Scientific Reports*, **2025**.
3. A. Kratimenos, K. Avramidis, C. Garoufis, A. Zlatintsi, and P. Maragos, “Augmentation Methods on Monophonic Audio for Instrument Classification in Polyphonic Music,” *Proceedings of the European Signal Processing Conference (EUSIPCO)*, **2020**. DOI: 10.23919/Eusipco47968.2020.9287745.



Proposed Method :

The audio will be sampled at 22,050 Hz and processed using Short-Time Fourier Transform (STFT) analysis to obtain a time-frequency representation. A Hann window, a 2,048-point transform, and a hop length of 512 samples will be used to generate Mel-spectrogram features with 128 Mel frequency bands.

- **Mel-spectrogram – represents the time-varying frequency content on a perceptually motivated Mel-frequency scale.**
- **STFT-based spectral features – describe how the frequency content of the audio changes over time and provide structured input for the CNN.**
- **Feature representation – each recording will be converted into a consistent Mel-spectrogram array suitable for model training and evaluation.**
- **Audio normalization and preparation – recordings will be processed consistently before feature extraction and classification.**

The main focus will be on Mel-spectrogram features, matching the processing pipeline used for the project. Other feature representations may be considered as future extensions rather than part of the core implementation. The extracted features will serve as input to the CNN models for learning discriminative patterns associated with different musical instrument classes. This representation captures the time-varying frequency characteristics of audio and provides a consistent input format for training and evaluating the classification models.

Machine Learning and Classification :

The extracted Mel-spectrograms will be supplied to CNN models. Stage 1 will treat recognition as an 11-class single-label task to predict the predominant instrument, while Stage 2 will use a multilabel model that can predict more than one instrument in an audio recording.

For Stage 2, synthetic mixtures of two instruments will be prepared for training, with instrument pairs represented in a balanced manner. The models will be evaluated using held-out data and standard classification metrics to assess their recognition performance.

The IRMAS dataset will be used for labelled musical recordings and evaluation. The multilabel model will also be tested on the official annotated IRMAS test recordings to examine how well it generalizes to real polyphonic audio.

Analysis and Explainability :

Model performance will be assessed by comparing predicted instrument labels with the available annotations. Evaluation will include accuracy for the single-label task and precision, recall, and F1-score for multilabel recognition.

A confusion matrix and class-wise performance measures will be used to examine strengths and weaknesses across instrument classes. The evaluation will also consider the challenges that arise when models trained on synthetic mixtures are applied to real recordings containing overlapping instrument sounds.

Expected Outcome :

The objective is to develop and evaluate a two-stage DSP- and ML-based musical instrument recognition system. The planned processing chain is:

Audio Signal → DSP Pre-processing → Mel-Spectrogram Extraction → Stage 1: Predominant-Instrument Classification → Stage 2: Multilabel Instrument Recognition → Performance Evaluation

The system will be evaluated using suitable measures, including accuracy, precision, recall, and F1-score, together with a confusion matrix for the single-label classifier. The project is intended to demonstrate how DSP-based audio representations and CNNs can be combined to recognize predominant and multiple instruments in musical recordings.

Tools/Software to be Used :

- **Python**
- **NumPy / SciPy** – signal processing
- **Librosa** – audio processing and feature extraction
- **Matplotlib** – waveform and spectrogram visualization
- **TensorFlow/Keras or PyTorch** – CNN/ML implementation
- **IRMAS dataset** – training and testing